

YLITORNIO MELTOSJARVI, Finland

WMO: 028120

Lat: 66.529N

Lon: 24.650E

Elev: 93

StdP: 100.21

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.7	-35.5	0.1	-31.9	-31.8	0.2	-28.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	26.2	17.0	23.9	16.1	21.9	15.0	18.6	23.2	17.4	22.1	16.2	20.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	20.3	15.7	11.3	18.4	14.7	10.5	17.3	53.2	23.2	49.1	22.3	45.8	20.2	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-34.4	28.2	3.6	2.7	-37.0	30.2	-39.1	31.8	-41.1	33.4	-43.7	35.4	(4)
(5)			WB	-34.5	20.3	3.5	1.6	-37.1	21.4	-39.2	22.3	-41.2	23.2	-43.7	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-13.1	-11.1	-6.2	0.4	7.4	12.1	15.4	12.7	8.3	0.8	-4.4	-8.4	(6)
(7)		DBStd	10.90	7.65	8.46	5.79	3.98	4.54	3.73	3.29	3.49	3.46	4.30	6.00	7.05	(7)
(8)		HDD10.0	3600	716	590	503	289	108	19	2	13	71	284	433	571	(8)
(9)		HDD18.3	6263	974	823	761	539	340	191	102	176	300	543	683	829	(9)
(10)		CDD10.0	397	0	0	0	0	27	82	168	99	21	0	0	0	(10)
(11)		CDD18.3	17	0	0	0	0	0	3	11	3	0	0	0	0	(11)
(12)		CDH23.3	251	0	0	0	0	21	47	153	29	0	0	0	0	(12)
(13)		CDH26.7	46	0	0	0	0	2	9	31	3	0	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	576	38	36	32	33	37	70	78	63	52	48	50	42	(15)
(16)		PrecMax	768	62	80	60	72	95	131	125	139	109	99	88	91	(16)
(17)		PrecMin	437	13	6	6	8	7	7	29	23	14	6	13	10	(17)
(18)		PrecStd	98	15	19	16	17	19	30	25	32	27	23	20	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	4.6	6.2	16.3	25.9	27.8	29.9	26.9	18.7	10.5	6.2	3.2	(19)
(20)			MCWB	0.7	2.8	2.6	8.3	13.8	17.7	19.3	19.8	13.8	8.7	5.7	2.7	(20)
(21)		2%	DB	0.0	2.0	4.2	11.0	22.7	24.3	27.2	23.6	17.0	9.4	4.5	1.5	(21)
(22)			MCWB	-0.2	1.1	1.3	5.3	13.8	15.5	17.5	16.7	12.5	8.0	4.0	1.3	(22)
(23)		5%	DB	-1.1	0.6	2.6	8.7	19.0	21.7	24.8	21.1	15.4	8.2	3.2	0.7	(23)
(24)			MCWB	-1.3	0.3	0.5	4.2	11.6	13.8	16.6	15.3	12.5	7.3	3.0	0.5	(24)
(25)		10%	DB	-2.7	-0.4	1.2	6.5	15.9	19.4	22.8	19.0	14.0	6.9	2.2	0.1	(25)
(26)			MCWB	-2.9	-0.6	0.0	2.8	9.5	12.8	16.1	14.6	11.7	6.0	2.0	0.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	2.8	3.1	8.4	15.5	18.5	20.7	20.0	15.0	9.8	6.0	2.7	(27)
(28)			MCDB	1.0	4.3	5.7	15.4	23.0	24.9	25.9	24.7	16.5	10.0	6.0	3.0	(28)
(29)		2%	WB	-0.2	1.2	1.8	5.9	13.9	16.9	19.1	17.8	13.8	8.2	4.1	1.3	(29)
(30)			MCDB	0.0	1.7	3.4	10.0	21.2	23.2	23.2	21.4	15.3	8.9	4.4	1.4	(30)
(31)		5%	WB	-1.3	0.3	1.0	4.5	12.3	15.5	18.1	16.4	13.0	7.3	3.0	0.6	(31)
(32)			MCDB	-1.1	0.7	1.9	8.2	18.2	19.5	23.2	19.4	14.6	8.1	3.1	0.7	(32)
(33)		10%	WB	-2.9	-0.7	0.3	3.2	10.5	13.9	16.9	15.4	12.0	6.1	2.0	0.0	(33)
(34)			MCDB	-2.7	-0.4	1.2	5.8	14.9	17.5	21.3	17.8	13.7	6.7	2.2	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	8.6	10.7	9.7	10.4	9.8	10.6	10.1	8.2	5.9	5.7	7.2	(35)
(36)			MCDBR	7.4	5.5	9.2	12.7	16.3	14.1	14.8	13.4	11.2	7.3	4.3	5.1	(36)
(37)			MCWBR	7.4	4.8	7.0	7.6	8.1	6.4	6.8	7.0	7.3	6.0	4.3	5.1	(37)
(38)		5% WB	MCDBR	7.5	5.4	6.6	12.1	14.7	11.7	11.9	10.1	8.2	6.3	4.3	4.9	(38)
(39)			MCWBR	7.5	4.8	5.3	7.5	7.6	6.0	6.4	5.9	5.7	5.5	4.3	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.199	0.267	0.298	0.333	0.343	0.346	0.359	0.342	0.317	0.290	0.345	0.360	(40)	
(41)		taud	1.938	2.108	2.178	2.253	2.449	2.469	2.450	2.481	2.536	2.351	2.645	2.680	(41)	
(42)		Ebn at Noon	202	570	751	823	859	864	837	815	757	561	215	53	(42)	
(43)		Edh at Noon	15	53	84	102	93	94	93	82	61	43	16	8	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.07	0.54	1.95	3.60	4.70	5.20	4.90	3.56	1.88	0.66	0.13	0.01	(44)	
(45)		RadStd	0.01	0.10	0.20	0.30	0.44	0.40	0.46	0.31	0.21	0.13	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page